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(54) **WATER HEATING SYSTEM AND
DETERMINATION METHOD OF TANK
CAPACITY IN WATER HEATING SYSTEM**

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(57) **ABSTRACT**
A water heating system includes a tank that retains water, a heat pump that heats the water in the tank, and a water heater that heats supplied water by combustion heat of fuel gas. In the water heating system, a tank use mode in which the water in the tank is mixed with tap water and supplied to a plurality of using locations including a bathtub, and a hot water supply use mode in which water supplied from the tank or tap water is heated by the water heater and then supplied to the plurality of using locations are executable. The tank use mode is executable at least in a predetermined second time slot, and a capacity of the tank is set to be equal to or more than a first reference capacity and equal to or less than a second reference capacity.

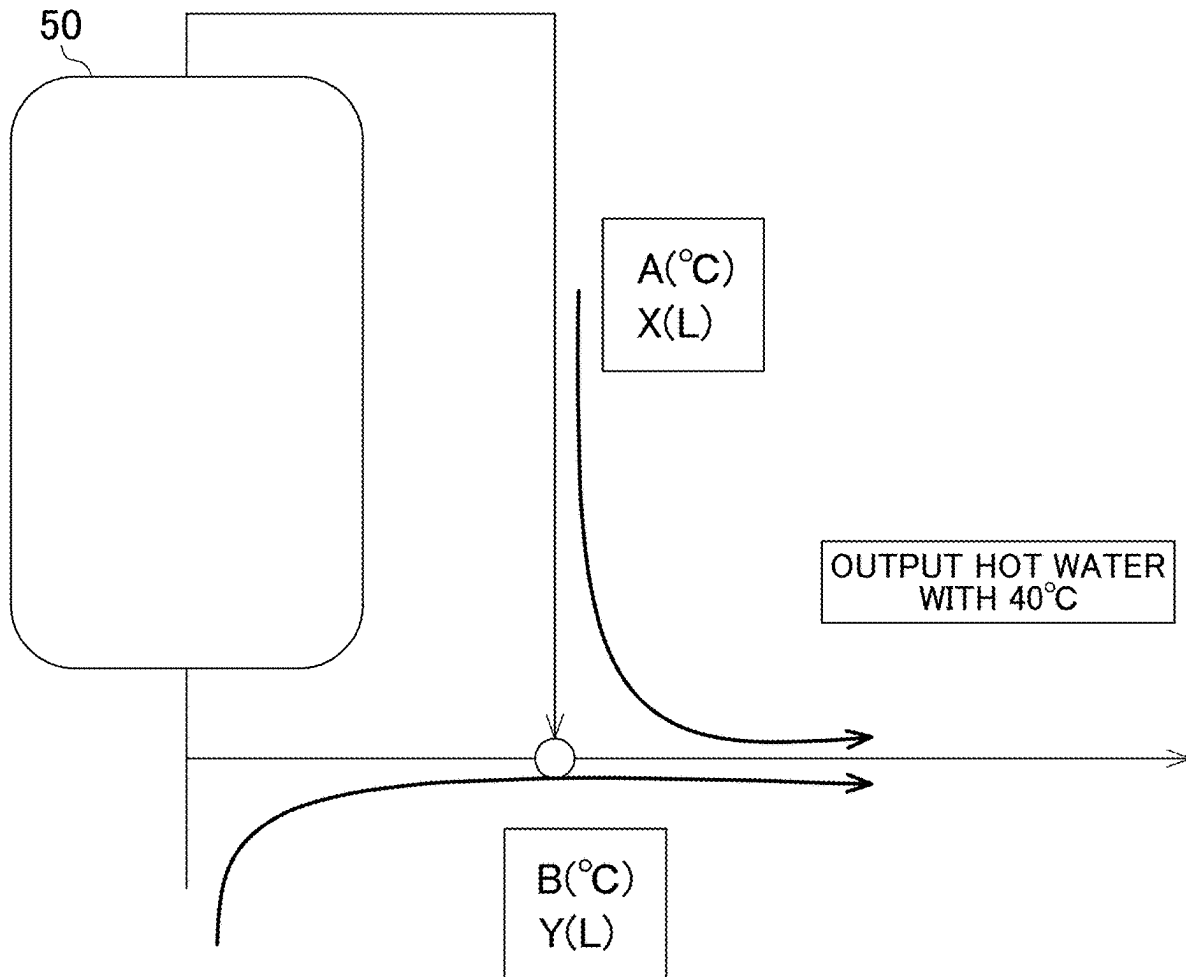


FIG.1A

TABLE SHOWING STARTING TIME OF HOT WATER SUPPLY AND THE LIKE,
OUTPUT HOT WATER AMOUNT, AND THE LIKE IN FOUR-PERSON
HOUSEHOLD CONSIDERED AS TYPICAL(REFER TO JIS 2075)

USAGE	STARTING TIME	PERIOD	INTERVAL	FLOW RATE	AMOUNT USED
		(s)	(s)	(L/m)	(L)
(1) FACE WASHING	6:45:00	120	—	5	10
	6:47:30	10	30	5	0.83
	6:48:10	10	30	5	0.83
	6:49:20	10	30	5	0.83
	6:50:00	10	30	5	0.83
					13.32
(2) KITCHEN	8:00:00	60	4190	5	5
	8:01:30	10	30	5	0.83
	8:02:10	10	30	5	0.83
	8:12:20	300	600	5	25
	8:19:20	30	120	5	2.5
					34.16
(3) KITCHEN	12:45:00	60	15910	5	5
	12:46:30	10	30	5	0.83
	12:47:10	10	30	5	0.83
	12:52:20	120	300	5	10
	12:55:20	30	60	5	2.5
					19.16
(4) KITCHEN	18:00:00	60	18250	5	5
	18:01:30	10	30	5	0.83
	18:03:40	60	120	5	5
	18:09:40	60	300	5	5
	18:11:10	10	30	5	0.83
	18:11:50	10	30	5	0.83
	18:12:30	10	30	5	0.83
	18:17:40	30	300	5	2.5
	18:18:40	10	30	5	0.83
	18:19:20	10	30	5	0.83
					22.48

FIG.1B

TABLE SHOWING STARTING TIME OF HOT WATER SUPPLY AND THE LIKE,
OUTPUT HOT WATER AMOUNT, AND THE LIKE IN FOUR-PERSON
HOUSEHOLD CONSIDERED AS TYPICAL(REFER TO JIS 2075)

(5)

USAGE	STARTING TIME	PERIOD	INTERVAL	FLOW RATE	AMOUNT USED
		(s)	(s)	(L/m)	(L)
BATH WATER FILLING	19:30:00	720	4230	15	180
KITCHEN	19:45:00	120	180	5	10
	19:47:30	30	30	5	2.5
SHOWER	19:53:00	120	300	10	20
KITCHEN	19:57:00	30	120	5	2.5
	19:58:00	10	30	5	0.83
	20:03:10	30	300	5	2.5
	20:04:10	10	30	5	0.83
	20:04:50	10	30	5	0.83
REHEATING	20:12:00				
SHOWER	20:15:00	300	600	10	50
KITCHEN	20:21:00	10	60	5	0.83
	20:21:40	10	30	5	0.83
	20:22:20	10	30	5	0.83
	20:23:00	10	30	5	0.83
REHEATING	20:42:00				
REHEATING	21:12:00				
REHEATING	21:42:00				
FACE WASHING	21:45:00	120	4910	5	10
	21:47:30	10	30	5	0.83
	21:48:10	10	30	5	0.83
	21:48:50	10	30	5	0.83
	21:59:00	120	600	10	20
	22:01:30	10	30	5	0.83
	22:02:10	10	30	5	0.83
REHEATING	22:12:00				
SHOWER	22:17:20	300	900	10	50
	22:24:20	30	120	5	2.5
	22:25:50	10	60	5	0.83
	22:28:00	60	120	5	5
	22:30:00	10	60	5	0.83

FIG.2

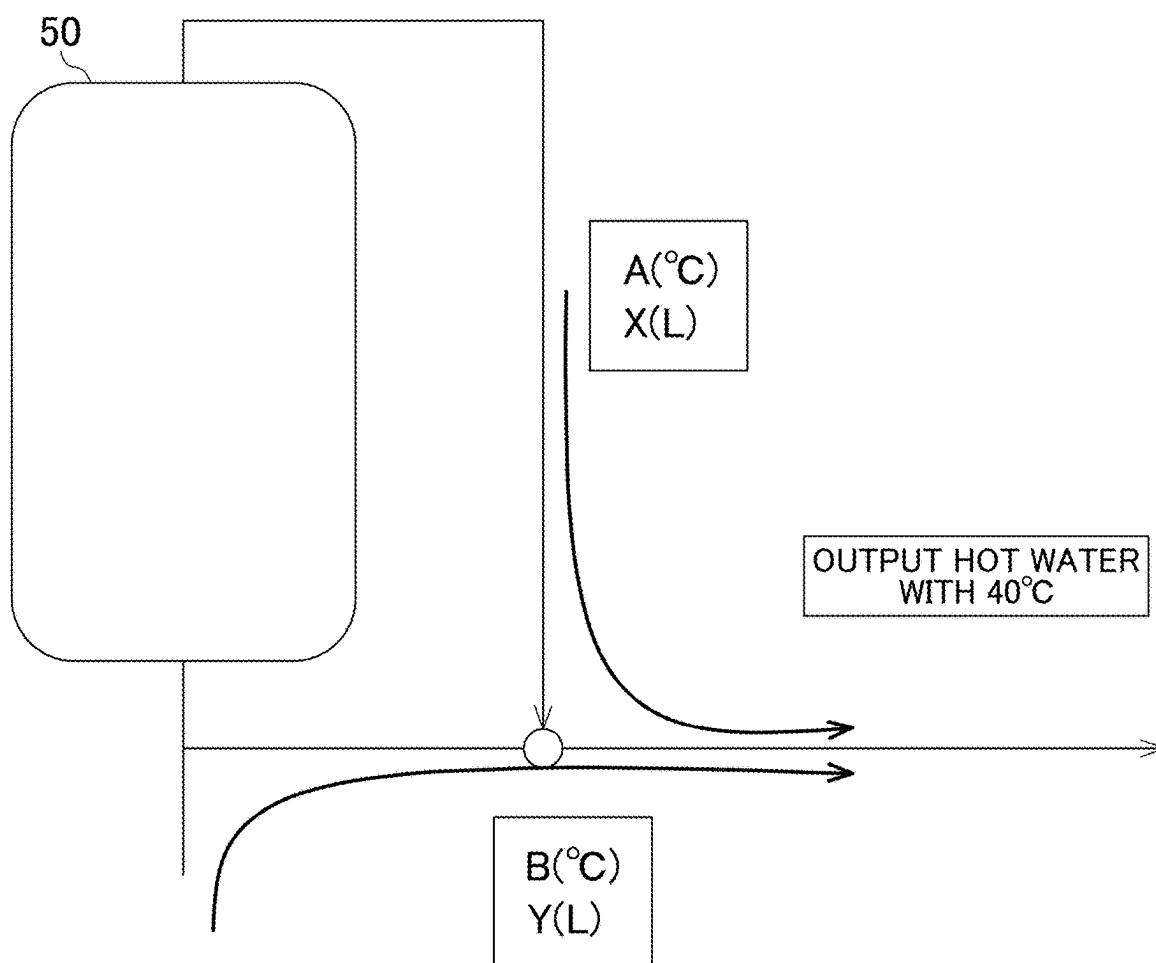


FIG.3

REQUIRED HOT WATER AMOUNT

INFLOW WATER TEMPERATURE TANK TEMPERATURE	50	55	60	65	70	75	80
5	26.56	23.91	21.74	19.92	18.39	17.08	15.94
7.6(LOWEST INFLOW WATER TEMPERATURE IN TOKYO)	26.10	23.34	21.12	19.28	17.73	16.42	15.29
10	25.62	22.77	20.50	18.63	17.08	15.77	14.66
15	24.40	21.35	19.00	17.08	15.53	14.23	13.14
16.8(+48.2°C)	23.87	20.74	18.35	16.44	14.89	13.62	12.54
20	22.77	19.52	17.08	15.18	13.66	12.42	11.39
25	20.50	17.08	14.64	12.81	11.39	10.25	9.32

FIG.4

CAPACITY THAT ALLOWS HEATING UP TO 65°C

INFLOW WATER TEMPERATURE	TANK TEMPERATURE	1.0Kw	1.5Kw	2.0Kw	2.5Kw	3.0Kw	3.5Kw	4.0Kw
5(+60°C)		14.33	21.50	28.67	35.83	46.90	50.17	57.33
7.6(+57.4°C)		14.98	22.47	29.96	37.46	44.94	52.43	59.93
10(+55°C)		14.33	21.50	28.67	35.83	49.60	54.73	62.54
15(+50°C)		15.64	23.45	31.27	39.09	51.60	60.20	68.80
16.8(+48.2°C)		17.80	26.80	36.70	44.60	53.53	62.45	71.37
20(+45°C)		19.11	28.67	38.22	47.78	57.33	66.89	76.44
25(+40°C)		21.50	32.30	43.00	53.80	64.50	75.25	86.00

FIG.5

CAPACITY THAT ALLOWS HEATING UP TO 75°C

INFLOW WATER TEMPERATURE	TANK TEMPERATURE		1.0Kw	1.5Kw	2.0Kw	2.5Kw	3.0Kw	3.5Kw	4.0Kw
5(+70°C)			12.29	18.43	24.57	30.71	39.69	46.31	52.90
7.6(+67.4°C)			12.76	19.14	25.52	31.90	41.34	48.24	55.13
10(+65°C)			13.23	19.85	26.46	33.08	43.00	50.17	57.33
15(+60°C)			14.33	21.50	28.67	35.83	46.90	54.73	62.54
16.8(+58.2°C)			14.78	22.16	29.56	36.94	48.50	56.58	64.66
20(+55°C)			15.64	23.45	31.27	39.09	51.60	60.20	68.80
25(+50°C)			17.20	25.80	34.40	43.00	57.33	66.89	76.44

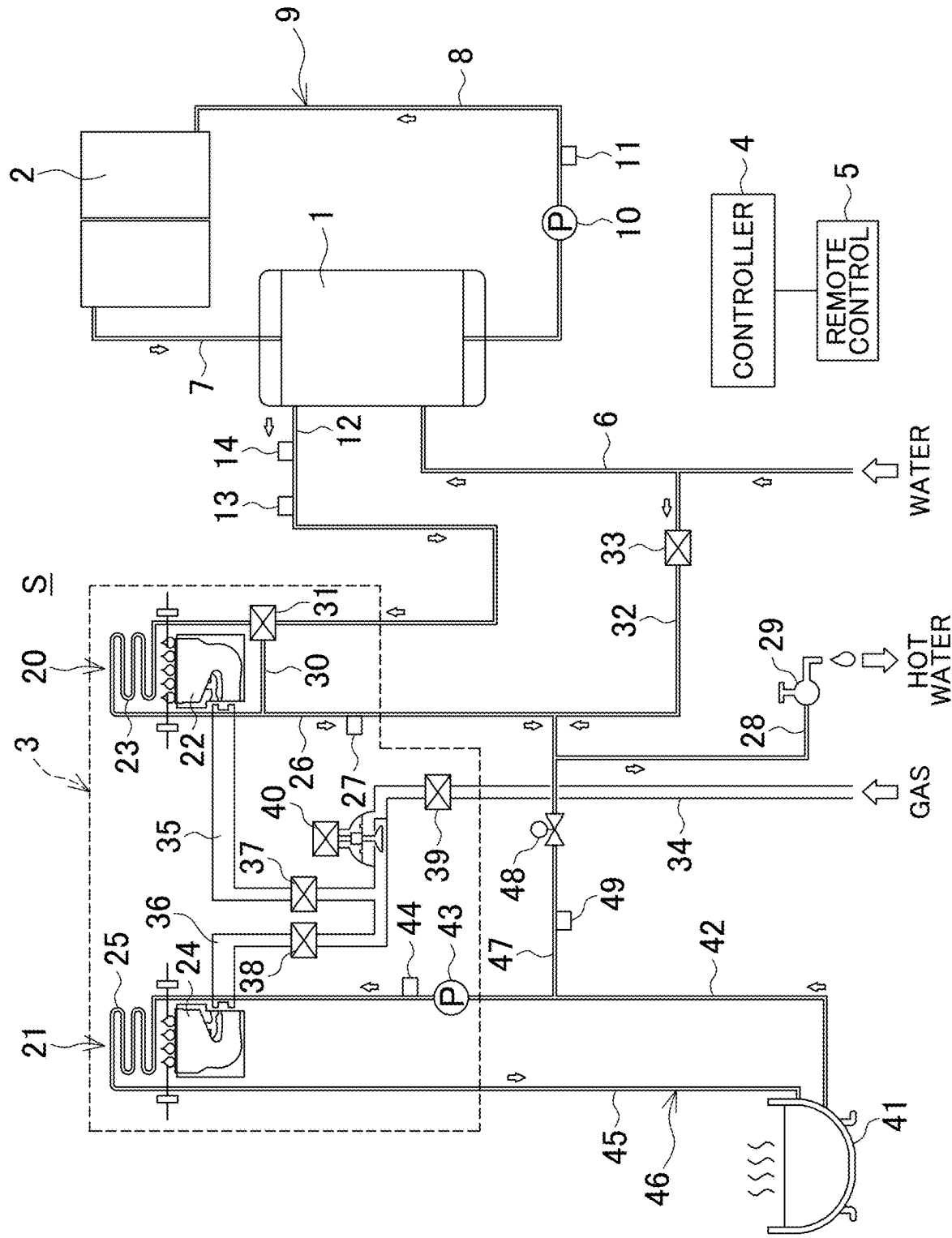


FIG.6

FIG. 7

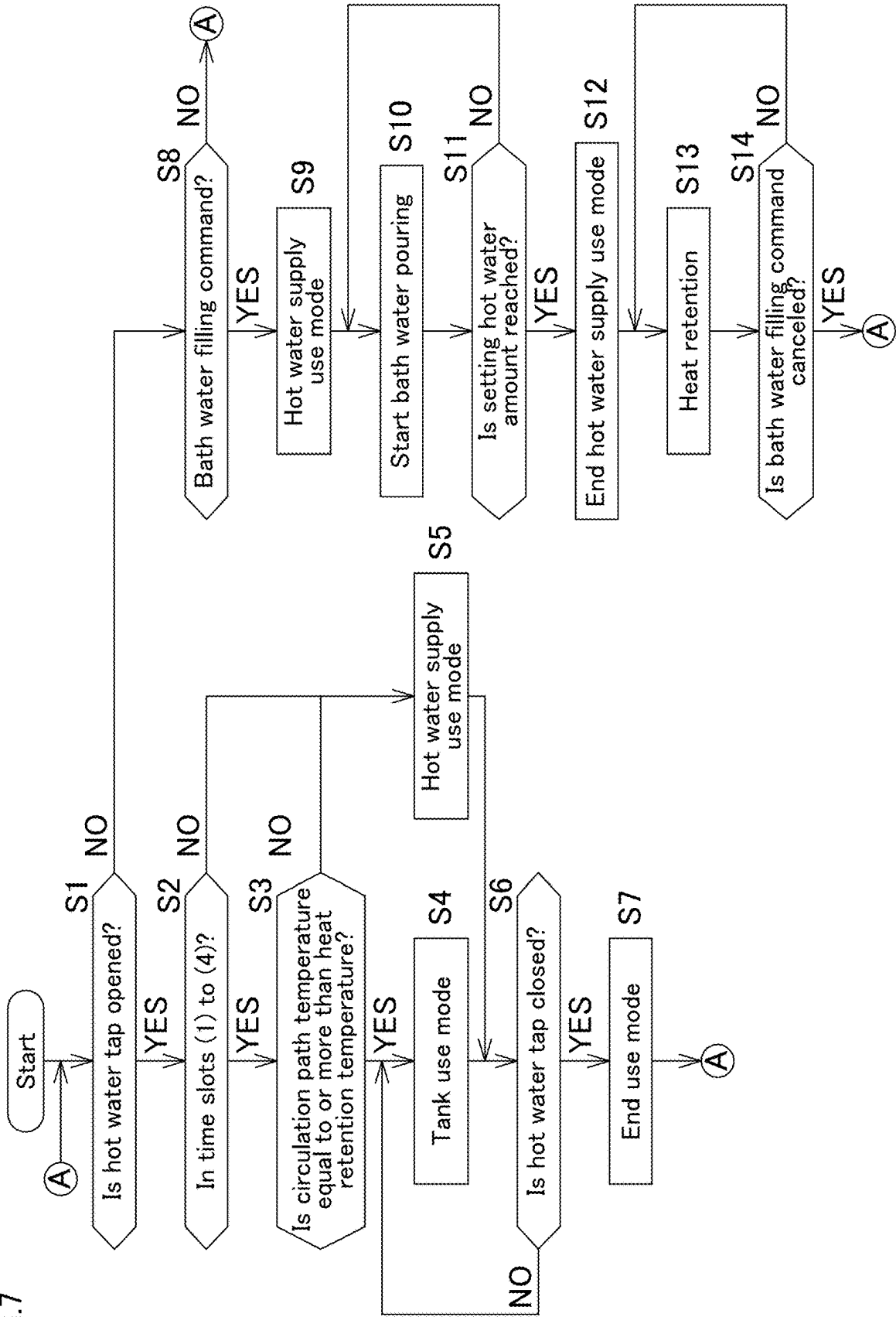


FIG.8A

TABLE III.1—POINT-OF-USE DRAW PATTERN

Draw No.	Time during test [hh:mm]	Volume [gallons (L)]	Flow rate ** [gpm (Lpm)]
1 *	0:00	2.0 (7.6)	1 (3.8)
2 *	1:00	1.0 (3.8)	1 (3.8)
3 *	1:05	0.5 (1.9)	1 (3.8)
4 *	1:10	0.5 (1.9)	1 (3.8)
5 *	1:15	0.5 (1.9)	1 (3.8)
6	8:00	1.0 (3.8)	1 (3.8)
7	8:15	2.0 (7.6)	1 (3.8)
8	9:00	1.5 (5.7)	1 (3.8)
9	9:15	1.0 (3.8)	1 (3.8)
Total Volume Drawn per Day: 10 gallons (38 L)			

* Denotes draws in first draw cluster.

** Should the water heater have a rated maximum gpm rating less than 1 gpm (3.8 Lpm), then all draws shall be implemented at a flow rate equal to the rated maximum gpm rating

FIG.8B

TABLE III.2—LOW-USAGE DRAW PATTERN

Draw No.	Time during test (hh:mm)	Volume (gallons)	Flow rate (gpm)
1*	0:00	15.0 (56.8)	1.7 (6.4)
2*	0:30	2.0 (7.6)	1 (3.8)
3*	1:00	1.0 (3.8)	1 (3.8)
4	10:30	6.0 (22.7)	1.7 (6.4)
5	11:30	4.0 (15.1)	1.7 (6.4)
6	12:00	1.0 (3.8)	1 (3.8)
7	12:45	1.0 (3.8)	1 (3.8)
8	12:50	1.0 (3.8)	1 (3.8)
9	16:15	2.0 (7.6)	1 (3.8)
10	16:45	2.0 (7.6)	1.7 (6.4)
11	17:00	3.0 (11.4)	1.7 (6.4)
Total Volume Drawn per Day: 38 gallons (144 L)			

* Denotes draws in first draw cluster.

FIG.8C

TABLE III.3—MEDIUM-USAGE DRAW PATTERN

Draw No.	Time during test (hh:mm)	Volume (gallons)	Flow rate (gpm)
1*	0:00	15.0 (56.8)	1.7 (6.4)
2*	0:30	2.0 (7.6)	1 (3.8)
3*	1:40	9.0 (34.1)	1.7 (6.4)
4	10:30	9.0 (34.1)	1.7 (6.4)
5	11:30	5.0 (18.9)	1.7 (6.4)
6	12:00	1.0 (3.8)	1 (3.8)
7	12:45	1.0 (3.8)	1 (3.8)
8	12:50	1.0 (3.8)	1 (3.8)
9	16:00	1.0 (3.8)	1 (3.8)
10	16:15	2.0 (7.6)	1 (3.8)
11	16:45	2.0 (7.6)	1.7 (6.4)
12	17:00	7.0 (26.5)	1.7 (6.4)
Total Volume Drawn Per Day: 55 gallons (208 L)			

* Denotes draws in first draw cluster.

WATER HEATING SYSTEM AND DETERMINATION METHOD OF TANK CAPACITY IN WATER HEATING SYSTEM

FIELD OF THE INVENTION

[0001] The disclosure relates to a hybrid-type water heating system that includes a water heater equipped with a burner and capable of heating passing water, and a tank equipped with a heat pump and retaining heated water. The disclosure also relates to a method for determining the capacity of the tank in the water heating system.

BACKGROUND OF THE INVENTION

[0002] Conventionally, a heat pump-type water heating system that includes a heat pump and a tank that retains water heated by the heat pump and supplies hot water in the tank to kitchens, bathrooms, and the like has been put into practical use.

[0003] In the water heating system, water (water including hot water) in the tank is heated by running the heat pump in time slots during which the amount of hot water used is relatively low, and when a hot water tap is opened or when a bath water filling command is issued, hot water is supplied from the tank. Accordingly, in order to supply a required amount of hot water to a bathtub, it is necessary to form the tank to be in a size that can retain a large amount of hot water.

[0004] Therefore, J P 2013-224762 A discloses a hybrid-type water heating system in which a burner heating device, which is a water heater, is installed in a supply passage on a downstream side of a tank that retains water heated by a heat pump, and the burner heating device is activated when the temperature of water supplied from the tank is lower than a hot water setting temperature or when a hot water amount in the tank is small during bath water filling.

[0005] With the above-described conventional hybrid-type water heating system, by using the water heater as an auxiliary heat source, the capacity of the tank can be relatively reduced, and the tank can be downsized to some extent.

[0006] However, considering the usability of a four-person household, which is considered as standard, there are limitations to downsizing, and the tank with a capacity of 100 liters (L) is adopted. In this case, it is necessary to increase the output of the heat pump in order to heat up 100 L of hot water for bath water filling, resulting in an increase in the size of the heat pump together with the tank. Therefore, it is difficult to install the water heating system in houses where the distance between a building and an exterior structure is narrow and collective housing. In view of this, further downsizing is required even for the hybrid-type water heating system.

[0007] Therefore, it is an object of the disclosure to provide a hybrid-type water heating system that can downsize a tank and a heat pump without causing inconvenience to users, and a method for determining a tank capacity in the water heating system.

SUMMARY OF THE INVENTION

[0008] In order to achieve the above-described object, the applicant first analyzed how hot water with 40° C. was used in a day in a four-person household considered as typical with reference to Japanese Industrial Standard (JIS) 2075.

[0009] JIS 2075 is a regulation concerning the measurement method of efficiency for domestic gas and oil water heater under standard usage mode, which is a method for measuring the efficiency of hot water equipment using a standard use mode reflecting the actual usage of hot water in general households. Therefore, the actual usage of hot water disclosed in JIS 2075 can be used as a standard model showing how water with 40° C. is used in a day in a four-person household considered as typical.

[0010] Then, as shown in FIGS. 1A and 1B, it was found that there were four time slots (1) to (4), namely a second time slot, and a time slot (5), namely a first time slot. The four time slots (1) to (4) are the time slots that have periods of no water use for 60 minutes or more before and after the time slots and during which water has not been used in a bathroom. The time slot (5) is the time slot during which the use of water in a bathroom has been recognized.

[0011] It should be noted that while JIS 2075 describes a standard use mode for equipment with automatic heat retention, a standard use mode for equipment without automatic heat retention (which performs manual reheating), and a standard use mode for equipment that does not perform reheating, the respective standard use modes have items, such as the amount used, interval, and flow rate, other than reheating, in common. In addition, as described in detail below, in the time slot (5), 180 L of water is used only for bath water filling, and since it is clear that the water in the tank is exhausted without taking into account the amount of water used for reheating, the items for reheating are left blank.

[0012] Then, it was found that the amount of water used extremely increased in the time slot (5) because an operation of bath water filling, for which the amount of water used was 180 L, was included in the time slot (5), whereas the maximum amount of water used was 34.16 L in the time slots (1) to (4).

[0013] Accordingly, the applicant considered that as long as the water heater as an auxiliary heat source was mainly used in the time slot (5) and the capacity of the tank of the heat pump was secured enough to supply water in the time slots (1) to (4), downsizing of the tank and the entire water heating system could be achieved.

[0014] However, if the capacity of the tank is too small, hot water runs out in the middle of using the water in the time slots (1) to (4), causing inconvenience to users. Conversely, if the capacity of the tank is too large, when water is used too much in the time slots (1) to (4), the temperature of water in the full tank may not rise sufficiently by the next time slot (5), still causing inconvenience to users.

[0015] Therefore, even if the tank is to be downsized, it is necessary to set lower limit and upper limit values of the capacity appropriately.

[0016] First, setting the lower limit value of the tank capacity, that is, a first reference capacity, is examined.

[0017] In FIG. 1A, since there was sufficient interval before and after each time slot in the time slots (1) to (4) that had no water use in a bathtub, it was determined that even if the water in the tank was exhausted in the time slots (1) to (4), hot water in the tank could be heated up by the next time slot.

[0018] Accordingly, it was considered that, by determining an amount of water, which could produce a hot water amount used in a time slot during which water was used most among the time slots (1) to (4) by mixing a predeter-

mined amount of tap water, and setting the amount as the lower limit value of the tank capacity, a hot water amount required in the time slots (1) to (4) could be secured.

[0019] Specifically, the hot water amount used most among the time slots (1) to (4) is 34.16 L in the time slot (2). This is set as a target water amount.

[0020] Then, as shown in FIG. 2, hot water amount X (L) with a hot water temperature A (° C.) in a tank 50 was mixed with tap water amount Y (L) with a water temperature B (° C.) to calculate how much hot water amount X in the tank 50 and how much tap water amount Y were required to output 34.16 (L) of water with 40° C. in accordance with JIS 2075 based on the following formulas (1) and (2). The specific gravity of water was assumed to be constant at all temperatures for the calculation.

$$X + Y = 34.16 \quad (1)$$

$$A \times X + B \times Y = 40 \times 34.16 \quad (2)$$

[0021] As a result, the hot water amount in a tank required for mixing with tap water, namely a required hot water amount, summarized according to the relationship between an inflow water temperature and a temperature of the hot water in the tank, namely a tank temperature, is as shown in the table in FIG. 3.

[0022] In FIG. 3, 7.6° C. is the lowest water temperature in 2020 announced by the Bureau of Waterworks, Tokyo Metropolitan Government, and 16.8° C. is the average temperature. The tank temperature of 75° C. is the thermal death point of *salmonella*. When hot water in the tank is kept warm under normal operation, it is assumed that the temperature of the hot water is raised to the thermal death point of *salmonella* and then retained at a temperature lower than the thermal death point. Therefore, in FIG. 3, each required hot water amount is calculated at the tank temperature in 5° C. increments in a range of 50° C. to 70° C., which is lower than 75° C., and the required hot water amount at the tank temperature of 75° C. and 80° C., which is higher than 75° C., is calculated for reference.

[0023] The examination focused on a case where the inflow water temperature water was 7.6° C. under the condition of supplying water without becoming too much or too little even in winter. Assuming that the tank temperature is maintained at 70° C., the required hot water amount is 17.73 L. Therefore, it is considered that the first reference capacity of the tank should be 17.73 L or more.

[0024] In addition, assuming that the tank temperature is maintained at 65° C., the required hot water amount is 19.28 L. In this case, although the tank capacity increases, it has the advantage that the frequency of operation of a heat pump can be reduced compared to the case where the hot water temperature is maintained at 70° C.

[0025] On the other hand, if the required hot water amount is set to a value smaller than 17.73 L, the hot water amount is sufficient except in the winter when the inflow water temperature is high. However, in the winter, the tank temperature must be maintained at 75° C. or 80° C., which is higher than 70° C., and the operating period of the heat pump becomes longer. This is undesirable considering the running cost.

[0026] Based on the above, it can be said that when downsizing of the water heating system is the first priority,

the required hot water amount is preferably set to 17.73 L or more. Furthermore, considering the reduction in energy consumption by reducing the frequency of operation and the operating period of the heat pump, it can be said that the first reference capacity is preferably set to 19.28 L or more.

[0027] Next, setting the upper limit value of the tank capacity, that is, a second reference capacity, is examined.

[0028] In a case where it is assumed that water in the tank is exhausted and the tank is filled with water of the same temperature as tap water, as long as the water in the tank can be heated up to a sufficient temperature by the next time slot during which hot water is used, the water can be supplied without causing inconvenience to users.

[0029] Therefore, tank volume (L), which can heat water up to a predetermined temperature (65° C., 75° C.) from a state where the water in the tank was replaced with water of the same temperature as tap water, was calculated based on the following formula (3).

$$\{\text{Tank volume} \times (\text{heating-up temperature} - \text{inflow water temperature}) / (\text{Heater capacity} \times 860)\} \times 60 = \text{heating-up period} \quad (3)$$

[0030] The rated power consumption (kcal/h) per 1.0 kW of the heater is 860. The heater capacity (kW) is set to a heat amount (1.0 kW, 1.5 kW, 2.0 kW, 2.5 kW) with a smaller output than those of commonly used household heat pump units (heating amount of 4.5 kW to 7.2 kW, refer to the Mitsubishi Electric Corporation website) in order to ensure downsizing the water heating system.

[0031] Concerning the multiplication of 60 (minutes) in the formula (3), since the minimum interval is 4190 seconds (1 hour 9 minutes and 50 seconds) between the time slot (1) and the time slot (2) among the intervals (non-use periods), during which water is not used after each of the time slots (1) to (4), in the table in FIG. 1A, the 60 (minutes) shorter than 4190 seconds is set.

[0032] Then, the capacity that allows heating up to 65° C. in 60 minutes was calculated according to the inflow water temperature and the heater capacity, and summarized in the table in FIG. 4. The capacity that allows heating up to 75° C. in 60 minutes was also calculated similarly and summarized in the table in FIG. 5. The output of the heater varies from 1.0 kW to 4.0 kW in 0.5 kW increments.

[0033] According to the table in FIG. 4, when the inflow water temperature is set to the above-described lowest inflow water temperature of 7.6° C., and the output of the heater is set to 2.0 kW as a small output, the capacity that allows heating up to 65° C. is 29.96 L. Specifically, considering the adoption of a heat pump with a small output (2.0 kW or less) to heat tap water in the tank up to 65° C. by the next time slot during which water is used, the second reference capacity of the tank should be 29.96 L or less. This is because, with a capacity more than 29.96 L, the tap water in the tank cannot be heated up to a sufficient temperature in 60 minutes, and therefore, the heat pump must have a high output, making it difficult to downsize the entire system.

[0034] In addition, considering that the water in the tank is heated up to the thermal death point of *salmonella* of 75° C., the second reference capacity of the tank is preferably set to 25.52 L or less, as shown in the table in FIG. 5. This is because, in this case, the tap water in the tank can be heated up to 75° C. in 60 minutes, and even by adding a period of one minute from heating up to sterilization, water of the required temperature can be prepared by the next time slot with plenty of time to spare.

[0035] Based on the above, a first configuration of the water heating system of the disclosure to achieve the above-described object has a configuration below.

[0036] A water heating system includes a tank that retains water, a heat pump that heats the water in the tank, and a water heater that heats supplied water by combustion heat of fuel gas. In the water heating system, a tank use mode in which water in the tank is mixed with tap water and supplied to a plurality of using locations including a bathtub, and a hot water supply use mode in which water supplied from the tank or tap water is heated by the water heater and then supplied to the plurality of using locations are executable, and the tank use mode is executable at least in a predetermined second time slot, excluding a predetermined first time slot in which bath water filling to the bathtub is assumed. An amount of hot water in a time slot during which hot water is used most is used as a target water amount. The time slot is among a plurality of time slots into which the second time slot is further subdivided. The plurality of subdivided time slots are assumed to have no water use for almost one hour or more before and after the time slots based on a daily usage condition of water in a general household considered as typical. An amount of water that allows producing mixed water with the target water amount of 40° C. is used as a first reference capacity, assuming that water between 65° C. and 75° C. is mixed with tap water supplied in winter. An amount of tap water whose temperature rises to a target temperature of 65° C. to 75° C. in one hour is used as a second reference capacity, assuming that the tap water is retained in the tank and heated with an output of the heat pump of 1.5 kw to 2.0 kw. A capacity of the tank is set to be equal to or more than the first reference capacity and equal to or less than the second reference capacity.

[0037] It should be noted that the first time slot is set, for example, including the above-described time slot (5), and the second time slot is set, for example, including the above-described time slots (1) to (4).

[0038] In another aspect of the first configuration, which is in the above-described configuration, the first reference capacity is 17.73 L, and the second reference capacity is 29.96 L.

[0039] In another aspect of the first configuration, which is in the above-described configuration, the first reference capacity is 19.28 L, and the second reference capacity is 25.52 L.

[0040] In another aspect of the first configuration, which is in the above-described configuration, the first reference capacity is a required hot water amount derived based on a relationship between an inflow water temperature of tap water and a temperature of water in the tank in order to output the target water amount at 40° C., and the target water amount is 34.16 L. The second reference capacity is a capacity that allows heating tap water in the tank up to a predetermined temperature by the heat pump within a predetermined non-use period during which water is not used in the second time slot, and the capacity is derived based on a relationship between the inflow water temperature of tap water and an output of the heat pump.

[0041] Furthermore, a second configuration of a method of determining a tank capacity in a water heating system of the disclosure to achieve the above-described object has a configuration below.

[0042] A method of determining a tank capacity in a water heating system is provided. The water heating system

includes a tank that retains water, a heat pump that heats the water in the tank, and a water heater that heats supplied water by combustion heat of fuel gas. The method determines a capacity of the tank in the water heating system. In the water heating system, a tank use mode in which water in the tank is mixed with tap water and supplied to a plurality of using locations including a bathtub, and a hot water supply use mode in which water supplied from the tank or tap water is heated by the water heater and then supplied to the plurality of using locations are executable, and the tank use mode is executable at least in a predetermined second time slot, excluding a predetermined first time slot in which bath water filling to the bathtub is assumed. The method includes using an amount of hot water in a time slot during which hot water is used most as a target water amount. The time slot is among a plurality of time slots into which the second time slot is further subdivided. The plurality of subdivided time slots are assumed to have no water use for almost one hour or more before and after the time slots based on a daily usage condition of water in a general household considered as typical. The method also includes using an amount of water that allows producing mixed water with the target water amount of 40° C. as a first reference capacity, assuming that water between 65° C. and 75° C. is mixed with tap water supplied in winter. The method also includes using an amount of tap water whose temperature rises to a target temperature of 65° C. to 75° C. in one hour as a second reference capacity, assuming that the tap water is retained in the tank and heated with an output of the heat pump of 1.5 kw to 2.0 kw. The method further includes setting a capacity of the tank to be equal to or more than the first reference capacity and equal to or less than the second reference capacity.

[0043] In another aspect of the second configuration, which is in the above-described configuration, the first reference capacity is 17.73 L, and the second reference capacity is 29.96 L.

[0044] In another aspect of the second configuration, which is in the above-described configuration, the first reference capacity is 19.28 L, and the second reference capacity is 25.52 L.

[0045] In another aspect of the second configuration, which is in the above-described configuration, the first reference capacity is a required hot water amount derived based on a relationship between an inflow water temperature of tap water and a temperature of water in the tank in order to output the target water amount at 40° C., and the target water amount is 34.16 L. The second reference capacity is a capacity that allows heating tap water in the tank up to a predetermined temperature by the heat pump within a predetermined non-use period during which water is not used in the second time slot, and the capacity is derived based on a relationship between the inflow water temperature of tap water and an output of the heat pump.

[0046] With the disclosure, since the capacity of the tank is set to be equal to or more than the first reference capacity and equal to or less than the second reference capacity, the tank and the heat pump can be downsized. Accordingly, the compactification of the entire water heating system can be achieved, and the water heating system can be installed with a small space.

[0047] In particular, the tank use mode is executable at least in the second time slot, excluding the first time slot in which bath water filling to a bathtub is assumed. Therefore,

the water in the tank can be used only in the time slots when the amount used is relatively low other than those with bath water filling, reducing the tank capacity as much as possible and reducing the possibility of running out of hot water even if the capacity is reduced. Therefore, inconvenience is not caused to users.

[0048] With another embodiment of the disclosure, in addition to the above effect, the first reference capacity is 17.73 L, leading to further downsizing of the tank and the heat pump.

[0049] With another embodiment of the disclosure, in addition to the above effects, by setting the first reference capacity to 19.28 L, the amount of hot water required in the above-described time slot (2) can be supplied even if a heat retention temperature in the tank is lowered to 65° C. Accordingly, the frequency of activation of the heat pump can be reduced to reduce the running cost.

[0050] In addition, by setting the second reference capacity to 25.52 L or less, even if the output of the heat pump is reduced to, for example, 2.0 kW, the temperature of the hot water in the tank can be raised to 75° C., which is the thermal death point of *salmonella*, in 60 minutes, making it easier to secure hygiene.

[0051] With another embodiment of the disclosure, in addition to the above effects, the first reference capacity is set to the required hot water amount derived based on the relationship between the inflow water temperature of tap water and the temperature of the water in the tank in order to output 34.16 L as the target water amount in the second time slot at 40° C., and the second reference capacity is the capacity derived based on the relationship between the inflow water temperature of tap water and the output of the heat pump as the capacity that can heat the tap water in the tank to a predetermined temperature by the heat pump within a predetermined non-use period during which water is not used in the second time slot. Therefore, even under a severe condition where the temperature of tap water becomes the lowest temperature (7.6° C.), 34.16 L, which is the amount used in the above-described time slot (2), of mixed hot water with 40° C. can be supplied.

BRIEF DESCRIPTION OF THE DRAWINGS

[0052] FIG. 1A is a table showing a starting time of hot water supply and the like, an output hot water amount, and the like in a four-person household considered as typical.

[0053] FIG. 1B is a table showing a starting time of hot water supply and the like, an output hot water amount, and the like in a four-person household considered as typical.

[0054] FIG. 2 is a schematic diagram of outputting hot water at a predetermined temperature of mixing water in a tank with tap water.

[0055] FIG. 3 is a table in which a hot water amount in the tank required for mixing with tap water, namely a required hot water amount, is summarized according to a relationship between an inflow water temperature and a temperature of the water in the tank, namely a tank temperature.

[0056] FIG. 4 is a table in which a capacity that allows heating up to 65° C. in 60 minutes is calculated for each the inflow water temperature and heater capacity and summarized.

[0057] FIG. 5 is a table in which a capacity that allows heating up to 75° C. in 60 minutes is calculated for each inflow water temperature and heater capacity and summarized.

[0058] FIG. 6 is a schematic diagram of a water heating system of an embodiment.

[0059] FIG. 7 is a flowchart of hot water supply control in the water heating system.

[0060] FIG. 8A is an excerpt of a part of the table of hot water usage pattern for each total volume used that is shown in the actual usage test of hot water described in the Federal Register of the United States.

[0061] FIG. 8B is an excerpt of a part of the table of hot water usage pattern for each total volume used that is shown in the actual usage test of hot water described in the Federal Register of the United States.

[0062] FIG. 8C is an excerpt of a part of the table of hot water usage pattern for each total volume used that is shown in the actual usage test of hot water described in the Federal Register of the United States.

DETAILED DESCRIPTION OF THE INVENTION

[0063] The following describes an embodiment of the disclosure based on the drawings. FIG. 6 is a schematic diagram illustrating an exemplary water heating system. A water heating system S includes a tank 1 that retains water, a heat pump 2 that heats the water in the tank 1, a water heater 3 that heats supplied water by combustion heat of fuel gas, a controller 4, and a remote control 5.

[0064] For a capacity of the tank 1 here, 20 L is selected based on the above-described determination method of a tank capacity. The heat pump 2 is a known heat source that includes an evaporator, a compressor, a heat exchanger, and the like and heats passing water through heat exchange between a heating medium and the passing water, and the heat pump 2 has an output of 2.0 kW. A water supply pipe 6 is connected to the tank 1. An external tap water pipe is connected to the water supply pipe 6.

[0065] A tank supply pipe 7 is connected between an upper portion of the tank 1 and an outlet of the heat pump 2. A tank return pipe 8 is connected between a lower portion of the tank 1 and an inlet of the heat pump 2. The tank supply pipe 7 and the tank return pipe 8 form a circulation path 9 through which water circulates between the heat pump 2 and the tank 1. The tank return pipe 8 includes a pump 10 and a return temperature thermistor 11 for detecting the temperature of water in the tank return pipe 8. A relay pipe 12 connected to a hot water supply heat exchanger 23 of the water heater 3 described later is connected to the upper portion of the tank 1. The relay pipe 12 includes a relay temperature thermistor 13 for detecting the temperature of water flowing through the relay pipe 12 and a flow rate sensor 14 for detecting the flow rate of water flowing through the relay pipe 12.

[0066] When heat retention operation with the heat pump 2 is instructed by the remote control 5, the controller 4 activates the heat pump 2 and the pump 10. Then, the water in the tank 1 circulates through the circulation path 9 and is heated when it passes through the heat pump 2. Once the water in the tank 1 is heated to a high temperature (for example, 75° C.), it is maintained at a predetermined heat retention temperature (for example, 65° C.) by controlling ON/OFF of the heat pump 2 based on the temperature detected by the return temperature thermistor 11.

[0067] The water heater 3 includes a hot water supply heating portion 20 and a bath heating portion 21. The hot water supply heating portion 20 has a hot water supply

burner 22 and a hot water supply heat exchanger 23. The bath heating portion 21 has a bath burner 24 and a bath heat exchanger 25.

[0068] In the hot water supply heating portion 20, the relay pipe 12 is connected to an inlet of the hot water supply heat exchanger 23. A hot water outlet pipe 26 is connected to an outlet of the hot water supply heat exchanger 23. An output hot water temperature thermistor 27 for detecting an output hot water temperature is disposed in the hot water outlet pipe 26. An external pipe 28 in which a plurality of hot water taps 29 are disposed is connected to the hot water outlet pipe 26. The hot water taps 29 are disposed in washrooms, kitchens, and bathrooms.

[0069] A bypass pipe 30 that bypasses the hot water supply heat exchanger 23 is connected between the relay pipe 12 and the hot water outlet pipe 26. A switching solenoid valve 31 is disposed at a connecting portion between the relay pipe 12 and the bypass pipe 30. The switching solenoid valve 31 allows the water flowing from the relay pipe 12 to switch to any of the hot water supply heat exchanger 23 side and the hot water outlet pipe 26 side.

[0070] A mixing pipe 32 is connected between the hot water outlet pipe 26 and the water supply pipe 6. The mixing pipe 32 includes a water amount control valve 33 that can control a water amount flowing from the water supply pipe 6 to the mixing pipe 32.

[0071] A gas pipe 34 that supplies fuel gas to the hot water supply burner 22 and the bath burner 24 is branched into a hot water supply branch pipe 35 on the hot water supply heating portion 20 side and a bath branch pipe 36 on the bath heating portion 21 side. The hot water supply branch pipe 35 supplies fuel gas to the hot water supply burner 22, and the bath branch pipe 36 supplies fuel gas to the bath burner 24. A hot water supply gas switching valve 37 is disposed in the hot water supply branch pipe 35, and a bath gas switching valve 38 is disposed in the bath branch pipe 36. In the gas pipe 34 before the branch, a main solenoid valve 39 and a proportional control valve 40 are disposed from an upstream side.

[0072] In the bath heating portion 21, a bath return pipe 42 is connected between an inlet of the bath heat exchanger 25 and an external bathtub 41. The bath return pipe 42 includes a circulation pump 43 and a bath temperature thermistor 44. A bath supply pipe 45 is connected between an outlet of the bath heat exchanger 25 and the bathtub 41. The bath return pipe 42 and the bath supply pipe 45 form a reheating circulation path 46 through which water circulates between the bath heat exchanger 25 and the bathtub 41. A bath water pouring pipe 47 is connected between the bath return pipe 42 and the hot water outlet pipe 26. The bath water pouring pipe 47 includes a bath water pouring solenoid valve 48 and a bath water pouring flow rate sensor 49.

[0073] The controller 4 is configured to include a CPU and a memory connected to the CPU and is electrically connected to each thermistor, each sensor, and each valve. The controller 4 executes heat retention operation for water in the tank 1, hot water supply operation to output hot water from the hot water taps 29, bath water filling operation to fill the bathtub 41 with water, and reheating operation to reheat the water in the bathtub 41. The controller 4 executes these operations in accordance with a program stored in a non-transitory computer-readable storage medium containing the memory connected to the CPU based on an operation

command and a setting temperature set by the remote control 5 and information obtained from each thermistor and each sensor.

[0074] However, in each operation, based on the time to use hot water shown in FIGS. 1A and 1B, a tank use mode in which water is supplied by mixing tap water with the water in the tank 1 and a hot water supply use mode in which water is supplied by heating the water supplied from the tank 1 or tap water by the water heater 3 are used separately depending on the time slot. The following describes the hot water supply control of the water heating system S by the controller 4 based on the flowchart in FIG. 7.

[0075] First, at Step (hereafter referred to simply as “S”) 1, when a hot water tap 29 is opened and passing water is confirmed by the flow rate sensor 14, the controller 4 determines whether or not a current time slot is a time slot from 12:00 a.m. to 7:00 p.m., including the time slots (1) to (4) in, FIG. 1A at S2.

[0076] When the current time slot is a time slot from 12:00 a.m. to 7:00 p.m., whether or not the temperature detected by the return temperature thermistor 11 is equal to or more than the heat retention temperature in the circulation path 9 is determined at S3.

[0077] Here, when it is confirmed to be equal to or more than the heat retention temperature, the tank use mode is executed at S4. Specifically, the controller 4 controls the switching solenoid valve 31 of the bypass pipe 30 to switch a flow passage of the relay pipe 12 to the bypass pipe 30 side. Then, tap water is supplied from the water supply pipe 6 into the tank 1, and water in the tank 1 flows from the relay pipe 12 through the bypass pipe 30 to the hot water outlet pipe 26 and output from the hot water tap 29. When the temperature of hot water obtained from the relay temperature thermistor 13 is higher than the setting temperature set by the remote control 5, the controller 4 controls the water amount control valve 33 of the mixing pipe 32 to a predetermined degree of opening to flow water from the water supply pipe 6 to the mixing pipe 32 and mix the water with the hot water in the hot water outlet pipe 26, thereby adjusting the temperature to the setting temperature.

[0078] On the other hand, when the current time slot is not a time slot from 12:00 a.m. to 7:00 p.m. including the time slots (1) to (4) in FIG. 1A at the determination of S2, that is, when the current time slot is a time slot from 7:00 p.m. to 12:00 a.m. including the time slot (5) in FIG. 1B, and when the temperature detected by the return temperature thermistor 11 falls below the heat retention temperature in the circulation path 9 at the determination of S3, the controller 4 executes the hot water supply use mode at S5. Specifically, fuel gas is supplied to the hot water supply burner 22 by controlling the switching solenoid valve 31 of the bypass pipe 30 to switch the flow passage of the relay pipe 12 to the hot water supply heat exchanger 23 side, opening the main solenoid valve 39 of the gas pipe 34 and the hot water supply gas switching valve 37, and opening the proportional control valve 40 at a predetermined degree of opening. An igniter is then activated to ignite the hot water supply burner 22. Then, tap water is supplied from the water supply pipe 6 into the tank 1, and water in the tank 1 passes through the hot water supply heat exchanger 23 from the relay pipe 12. Accordingly, the water is heated through heat exchange with combustion exhaust gas, flows to the hot water outlet pipe 26, and is output from the hot water tap 29.

[0079] In both the tank use mode of S4 and the hot water supply use mode of S5, when it is confirmed at S6 that the passing water has stopped due to a closure of the hot water tap 29, the controller 4 ends the current use mode (extinguishes the hot water supply burner 22 in the hot water supply use mode) at S7 and returns to S1.

[0080] The time slot from 7:00 p.m. to 12:00 a.m. is an example of a first time slot of the disclosure, and the time slot from 12:00 a.m. to 7:00 p.m. is an example of a second time slot of the disclosure.

[0081] When the passing water due to the opening of the hot water tap 29 is not confirmed at S1, and a bath water filling command is issued by the remote control 5 at S8, the controller 4 executes the hot water supply use mode at S9. Specifically, fuel gas is supplied to the hot water supply burner 22 by controlling the switching solenoid valve 31 of the bypass pipe 30 to switch the flow passage of the relay pipe 12 to the hot water supply heat exchanger 23 side, opening the main solenoid valve 39 of the gas pipe 34 and the hot water supply gas switching valve 37, and opening the proportional control valve 40 at a predetermined degree of opening. The igniter is then activated to ignite the hot water supply burner 22. Accordingly, water is heated through heat exchange with combustion exhaust gas and flows to the hot water outlet pipe 26.

[0082] Next, the controller 4 starts bath water pouring to the bathtub 41 by opening the bath water pouring solenoid valve 48 of the bath water pouring pipe 47 at S10.

[0083] Accordingly, the hot water heated by passing through the hot water supply heat exchanger 23 flows from the hot water outlet pipe 26 to the bath water pouring pipe 47 and is supplied from the bath return pipe 42 to the bathtub 41.

[0084] When it is confirmed at S11 that the flow rate obtained from the bath water pouring flow rate sensor 49 has reached a preset setting hot water amount, the controller 4 stops the combustion of the hot water supply burner 22 by closing the bath water pouring solenoid valve 48 at S12 and ends the hot water supply use mode.

[0085] After bath water filling, the hot water in the bathtub 41 is kept warm at S13. Specifically, the hot water temperature is monitored by the bath temperature thermistor 44, and when the hot water temperature falls below a set bath water filling temperature, reheating is performed by activating the circulation pump 43, opening the bath gas switching valve 38 to ignite the bath burner 24, and circulating the hot water in the bathtub 41 through the reheating circulation path 46. When the bath water filling command is canceled by the remote control 5 at S14, the control returns to S1.

[0086] The water heating system S of the above-described configuration includes the tank 1 that retains water, the heat pump 2 that heats the water in the tank 1, and the water heater 3 that heats the supplied water by combustion heat of fuel gas. In the water heating system S, the tank use mode in which the water in the tank 1 is mixed with tap water and supplied to a plurality of using locations, and the hot water supply use mode in which the water supplied from the tank 1 is heated by the water heater 3 and then supplied to the plurality of using locations are executable.

[0087] Since the capacity of the tank 1 is set to 20 L that is equal to or more than the first reference capacity and equal to or less than the second reference capacity based on the above-described determination method of the tank capacity, the tank 1 and the heat pump 2 can be downsized. Accord-

ingly, the compactification of the entire water heating system S can be achieved, and the water heating system S can be installed with a small space.

[0088] In particular, the tank use mode is assumed to be executed in the second time slot, excluding the first time slot from 7:00 p.m. to 12:00 a.m. that includes the above-described time slot (5) in which bath water filling to the bathtub 41 is assumed. Therefore, the water in the tank 1 can be used only in the time slots other than those with bath water filling when the amount used is relatively low, reducing the capacity of the tank 1 as much as possible and reducing the possibility of running out of hot water even if the capacity is reduced. Therefore, inconvenience is not caused to users.

[0089] Since the output of the heat pump 2 is sufficient at 2.0 kW, the heat pump 2 can also be downsized, leading to a further compactification of the entire water heating system S.

[0090] The following describes a modification example of the disclosure.

[0091] The capacity of the tank is not limited to 20 L of the above-described configuration, and as described above, the capacity may be set as appropriate between 17.73 L and 29.96 L, preferably between 19.28 L and 25.52 L. In the above-described configuration, the bypass pipe is connected between the relay pipe and the hot water outlet pipe, and in the tank use mode in which water in the tank is used, the water flows from the relay pipe to the bypass pipe. However, the bypass pipe may be eliminated, and in the tank use mode, water may flow from the relay pipe to the hot water supply heat exchanger.

[0092] Heat exchangers of the water heater may include a primary heat exchanger that recovers sensible heat and a secondary heat exchanger that recovers latent heat. While the water heater of the above-described configuration includes the bath heating portion, the water heater may only be able to perform bath water filling with the bath water pouring pipe without having the bath heating portion.

[0093] The first and second time slots are not fixed and can be changed accordingly, for example, by setting and operating the remote control. It is only necessary to set the first and second time slots according to the climate, environment, and the like where the water heating system is installed.

[0094] The usage condition of water in general households is not limited to the assumption based on the JIS standard, as in the above-described configuration, but may be assumed based on general foreign standards, such as the US standard.

[0095] The tank use mode is not limited to the case where it is executed only in the second time slot, as in the above-described configuration, but may be executed in the first time slot as long as the remaining amount of water in the tank is sufficient and used in a small amount and for a short period of time, for example, used for adding hot water to a bathtub.

[0096] In addition, the controller may store the amount of water used at each using location for a certain period, such as a week, ten days, or one month, and change the first and second time slots for executing the tank use mode and the hot water supply use mode based on the amount used during the stored certain period. Specifically, it is possible to, for example, change the second time slot to the first time slot at the timing when the average or accumulated value of the amount used during a certain period exceeds a predetermined threshold, or conversely, to change the first time slot

to the second time slot at the timing when the average or accumulated value of the amount used during a certain period falls below a predetermined threshold.

[0097] Therefore, the first time slot and the second time slot are not limited to the case where they are divided into two, such as the time slot including the above-described (1) to (4) and the time slot including the above-described (5), but may be a case where the first time slot and the second time slot are divided into three or more time slots to alternately execute the tank use mode and the hot water supply use mode.

[0098] In the above-described configuration, the capacity of the tank is determined based on the actual usage of hot water in general households in the Japan's JIS standard. However, in each country in which the water heating system of the disclosure is applied, the capacity of the tank may be determined by similarly creating a model case based on the actual usage of hot water in each country.

[0099] For example, in the United States, the Federal Register No. 78 published by the Department of Energy describes the fact that an actual usage test of water heaters was conducted for 24 hours to verify the elapsed time and the usage pattern (flow rate) of hot water for each total volume used. FIGS. 8A to 8C are excerpts (citations) of a part of the tables of hot water usage patterns for each total volume used that is shown in the Federal Register.

[0100] In FIG. 8C, the usage pattern under moderate total volume used (208 L) (TABLE III.3-MEDIUM-USAGE DRAW PATTERN) shows that the amount used immediately after the start of the test increases at 56.8 L assuming the use in bathrooms, but the maximum amount is 34.1 L after a lapse of 1 hour 40 minutes and after a lapse of 10 hours 30 minutes after the start of the test.

[0101] From here, it can be found that by determining the tank capacity using 34.1 L as a guideline for households with moderate total volume used, hot water can be supplied from a water heater during bath time when the amount of hot water used increases and can be supplied by heating water in the tank by a heat pump at other times.

[0102] Thus, in each country, it is only necessary create a model case based on the actual usage and the like and determine the tank capacity based on the model case.

[0103] It is explicitly stated that all features disclosed in the description and/or the claims are intended to be disclosed separately and independently from each other for the purpose of original disclosure as well as for the purpose of restricting the claimed invention independent of the composition of the features in the embodiments and/or the claims. It is explicitly stated that all value ranges or indications of groups of entities disclose every possible intermediate value or intermediate entity for the purpose of original disclosure as well as for the purpose of restricting the claimed invention, in particular as limits of value ranges.

1. A water heating system comprising:

- a tank that retains water;
 - a heat pump that heats the water in the tank; and
 - a water heater that heats supplied water by combustion heat of fuel gas, wherein
- in the water heating system,
- a tank use mode in which the water in the tank is mixed with tap water and supplied to a plurality of using locations including a bathtub, and
 - a hot water supply use mode in which water supplied from the tank or tap water is heated by the water

heater and then supplied to the plurality of using locations are executable, wherein

the tank use mode is executable at least in a predetermined second time slot, excluding a predetermined first time slot in which bath water filling to the bathtub is assumed, and

an amount of hot water in a time slot during which hot water is used most is used as a target water amount, the time slot is among a plurality of time slots into which the second time slot is further subdivided, the plurality of subdivided time slots are assumed to have no water use for almost one hour or more before and after the time slots based on a daily usage condition of water in a general household considered as typical,

an amount of water that allows producing mixed water with the target water amount of 40° C. is used as a first reference capacity, assuming that water between 65° C. and 75° C. is mixed with tap water supplied in winter,

an amount of tap water whose temperature rises to a target temperature of 65° C. to 75° C. in one hour is used as a second reference capacity, assuming that the tap water is retained in the tank and heated with an output of the heat pump of 1.5 kw to 2.0 kw, and

a capacity of the tank is set to be equal to or more than the first reference capacity and equal to or less than the second reference capacity.

2. The water heating system according to claim 1, wherein the first reference capacity is 17.73 L, and the second reference capacity is 29.96 L.

3. The water heating system according to claim 2, wherein the first reference capacity is 19.28 L, and the second reference capacity is 25.52 L.

4. The water heating system according to claim 1, wherein the first reference capacity is a required hot water amount derived based on a relationship between an inflow water temperature of tap water and a temperature of the water in the tank in order to output the target water amount at 40° C., and the target water amount is 34.16 L, and

the second reference capacity is a capacity that allows heating tap water in the tank up to a predetermined temperature by the heat pump within a predetermined non-use period during which water is not used in the second time slot, and the capacity is derived based on a relationship between the inflow water temperature of tap water and an output of the heat pump.

5. A method of determining a tank capacity in a water heating system, the water heating system comprising:

- a tank that retains water;
- a heat pump that heats the water in the tank; and
- a water heater that heats supplied water by combustion heat of fuel gas, wherein

the method determines a capacity of the tank in the water heating system,

in the water heating system,

a tank use mode in which the water in the tank is mixed with tap water and supplied to a plurality of using locations including a bathtub, and

a hot water supply use mode in which water supplied from the tank or tap water is heated by the water heater and then supplied to the plurality of using locations

are executable, wherein

the tank use mode is executable at least in a predetermined second time slot, excluding a predetermined first time slot in which bath water filling to the bathtub is assumed, wherein

the method comprises:

using an amount of hot water in a time slot during which hot water is used most as a target water amount, the time slot being among a plurality of time slots into which the second time slot is further subdivided, the plurality of subdivided time slots being assumed to have no water use for almost one hour or more before and after the time slots based on a daily usage condition of water in a general household considered as typical;

using an amount of water that allows producing mixed water with the target water amount of 40° C. in as a first reference capacity, assuming that water between 65° C. and 75° C. is mixed with tap water supplied in winter;

using an amount of tap water whose temperature rises to a target temperature of 65° C. to 75° C. in one hour as a second reference capacity, assuming that the tap water is retained in the tank and heated with an output of the heat pump of 1.5 kw to 2.0 kw; and

setting a capacity of the tank to be equal to or more than the first reference capacity and equal to or less than the second reference capacity.

6. The determination method of a tank capacity in a water heating system according to claim 5, wherein the first reference capacity is 17.73 L, and the second reference capacity is 29.96 L.

7. The determination method of a tank capacity in a water heating system according to claim 6, wherein the first reference capacity is 19.28 L, and the second reference capacity is 25.52 L.

8. The determination method of a tank capacity in a water heating system according to claim 5, wherein the first reference capacity is a required hot water amount derived based on a relationship between an inflow water temperature of tap water and a temperature of the water in the tank in order to output the target water amount at 40° C., and the target water amount is 34.16 L, and

the second reference capacity is a capacity that allows heating tap water in the tank up to a predetermined temperature by the heat pump within a predetermined non-use period during which water is not used in the second time slot, and the capacity is derived based on a relationship between the inflow water temperature of tap water and an output of the heat pump.

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